The documentation and process conversion measures necessary to comply with this document shall be completed by 7 May 2010.

INCH-POUND

MIL-PRF-19500/182G
w/AMENDMENT 1
7 February 2010
SUPERSEDING
MIL-PRF-19500/182G
2 August 2006

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, LOW-POWER,
TYPES 2N720A, 2N720AUB, 2N1893, AND 2N1893S, JAN, JANTX, JANTXV,
JANHC2N720A, AND JANKC2N720A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for for NPN silicon, low-power transistors. Three levels of product assurance are provided for each device type as specified in MIL-PRF-19500, two levels of product assurance are provided for die.

1.2 Physical dimensions. See figure 1 (similar to TO-18) figure 2 (similar to TO-5), figure 3 (UB ), and figure 4 (JANHC and JANKC die layout).

* 1.3 Maximum ratings unless otherwise specified $T_A$ = +25°C.

| Type | $P_{T1}$ (1) (2) $T_C$ = +25°C | $P_{T2}$ (3) (4) $T_A$ = +25°C | $P_{T3}$ (3) (4) $T_{SP}$ = +25°C | $R_{\theta JA}$ | $R_{\theta JC}$ | $R_{\theta JSP(IS)}$ | $V_{CBO}$ | $V_{CEO}$ | $V_{EBO}$ | $I_C$ | $T_J$ and $T_{STG}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | W | W | W | °C/W | °C/W | °C/W | V dc | V dc | V dc | mA dc | °C |
| 2N720A | 1.8 | 0.5 | | 325 | 97 | | 120 | 80 | 7 | 500 | -65 to |
| 2N720AUB | 1.16 | | 0.5 | 325 | | 150 | 120 | 80 | 7 | 500 | +200 |
| 2N1893 | 3.0 | 0.8 | | 175 | 58 | | 120 | 80 | 7 | 500 | |
| 2N1893S | 3.0 | 0.8 | | 175 | 58 | | 120 | 80 | 7 | 500 | |

* (1) Derate linearly at 10.3 mW/°C for type 2N720A, 6.63 mW/°C for type 2N720AUB and see figure 5 (TO-5) for type 2N1893 and 2N1893S for $T_C$ > +25°C.
* (2) For thermal impedance curve, for 2N1893 and 2N1893S see figure 6 (TO-18) for $T_C$ > +25°C.
* (3) Derate linearly at 3.08 mW/°C for types 2N720A and 2N720AUB; $T_A$ > +37.5°C and see figure 7 (TO-5) for types 2N1893 and 2N1893S for $T_A$ > +60°C.
* (4) For thermal impedance curve, for 2N1893 and 2N1893S see figure 8 (TO-18) for $T_A$ > +25°C..

* Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil .



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1.4 Primary electrical characteristics.

| Limits | $h_{FE1}$ (1) | $h_{FE2}$ (1) | $h_{FE3}$ (1) | $\lvert h_{fe} \rvert$ | $V_{CE(SAT)}$ (1) |
|---|---|---|---|---|---|
| | $V_{ce}$ = 10 V dc<br>$I_C$ = 0.1 mA dc | $V_{ce}$ = 10 V dc<br>$I_C$ = 10 mA dc | $V_{ce}$ = 10 V dc<br>$I_C$ = 150 mA dc | f = 20 MHz<br>$V_{ce}$ = 10 V dc<br>$I_C$ = 50 mA dc | $I_C$ = 50 mA dc<br>$I_B$ = 5.0 mA dc |
| Min<br>Max | 20 | 35 | 40<br>120 | 3.0<br>10 | V dc<br><br>1.2 |

(1) Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

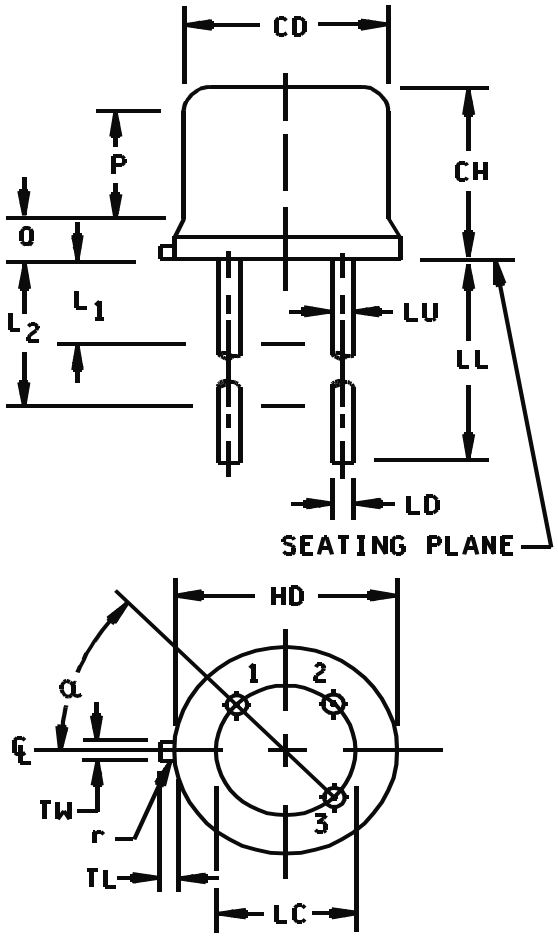
* (Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch or https://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

* 2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.



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| Ltr | Dimensions | | | | Notes |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| CD | .178 | .195 | 4.52 | 4.95 | |
| CH | .170 | .210 | 4.32 | 5.33 | |
| HD | .209 | .230 | 5.31 | 5.84 | |
| LC | .100 TP | | 2.54 TP | | 5 |
| LD | | .021 | | 0.53 | 3, 8 |
| LL | .500 | | 12.70 | | 8, 9 |
| LU | .016 | .019 | 0.41 | 0.48 | 3, 8, 9 |
| $L_1$ | | .050 | | 1.27 | 9 |
| $L_2$ | .250 | | 6.35 | | 9 |
| P | .100 | | 2.54 | | |
| Q | | .030 | | 0.76 | 4 |
| TL | .028 | .048 | 0.71 | 1.22 | 7 |
| TW | .036 | .046 | 0.91 | 1.17 | |
| r | | .010 | | 0.25 | |
| α | 45° TP | | 45° TP | | |



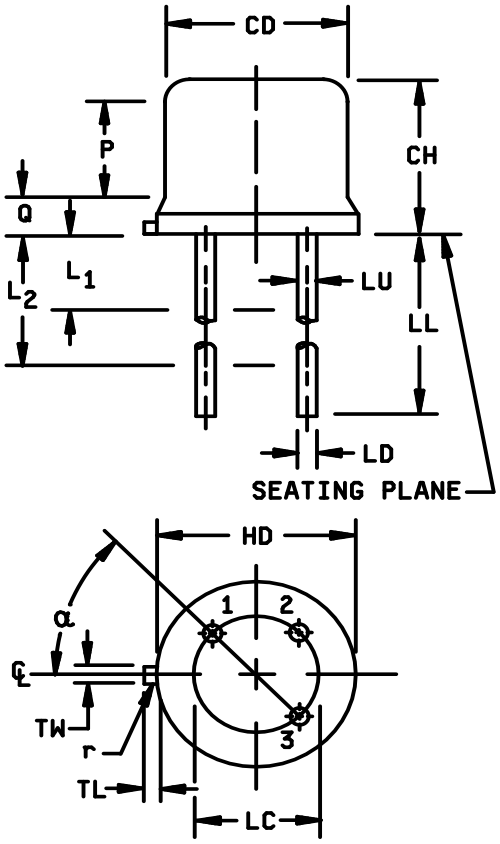

NOTES:
1. Dimensions are in inches.
2. Millimeters equivalents are given for general information only.
3. Measured in the zone beyond .250 inch (6.35 mm) from the seating plane.
4. Details of outline in this zone are optional.
5. When measured in a gauging plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below the seating plane of the transistor, maximum diameter leads shall be within .007 inch (0.18 mm) of their true location relative to a maximum width tab. Smaller diameter leads shall fall within the outline of the maximum diameter lead tolerance.
6. The collector shall be internally connected to the case.
7. Measured from the maximum diameter of the actual device.
8. All three leads.
9. Symbol LU applies between $L_1$ and $L_2$. Dimension LD applies between $L_2$ and LL minimum. Lead diameter shall not exceed .042 inch (1.07 mm) within $L_1$ and beyond LL minimum.
10. Lead 1 = emitter, lead 2 = base, lead 3 = collector.
11. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions for device type 2N720A (TO-18).



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| Dimensions | | | | | |
|---|---|---|---|---|---|
| Symbol | Inches | | Millimeters | | Notes |
| | Min | Max | Min | Max | |
| CD | .305 | .335 | 7.75 | 8.51 | |
| CH | .240 | .260 | 6.10 | 6.60 | |
| HD | .335 | .370 | 8.51 | 9.40 | |
| LC | .200 TP | | 5.08 TP | | 6 |
| LD | .016 | .021 | 0.41 | 0.53 | 7, 8 |
| LL | See notes | | | | 7, 8, 11,12 |
| LU | .016 | .019 | 0.41 | 0.48 | 7, 8 |
| $L_1$ | | .050 | | 1.27 | 7, 8 |
| $L_2$ | .250 | | 6.35 | | 7, 8 |
| P | .100 | | 2.54 | | 5 |
| Q | | .050 | | 1.27 | 4 |
| r | | .010 | | 0.25 | 10 |
| TL | .029 | .045 | 0.74 | 1.14 | 3 |
| TW | .028 | .034 | 0.71 | 0.86 | 2 |
| $\alpha$ | 45° TP | | 45° TP | | 6 |

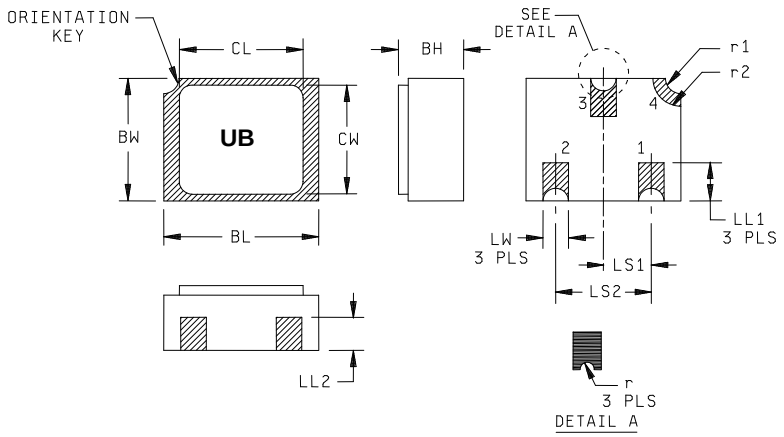


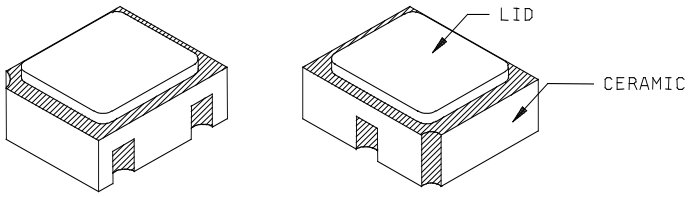

NOTES:
1. Dimension are in inches.
2. Millimeters are given for general information only.
3. Beyond r (radius) maximum, TH shall be held for a minimum length of .011 (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Body contour optional within zone defined by HD, CD, and Q.
6. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC.
7. Dimension LU applies between $L_1$ and $L_2$. Dimension LD applies between $L_2$ and LL minimum. Diameter is uncontrolled in $L_1$ and beyond LL minimum.
8. All three leads.
9. The collector shall be internally connected to the case.
10. Dimension r (radius) applies to both inside corners of tab.
11. For 2N1893, dimension LL shall be 1.5 inches (38.1mm) minimum and 1.75 inches (44.4 mm) maximum.
12. For 2N1893S, dimension LL shall be .5 inch (12.7mm) minimum and .75 inch (19.0 mm) maximum.
13. In accordance with ASME Y14.5M, diameters are equivalent to $\phi$x symbology.
14. Lead 1 = emitter, lead 2 = base, lead 3 = collector.

FIGURE 2. Physical dimensions for device types 2N1893 and 2N1893S (similar to TO-5).



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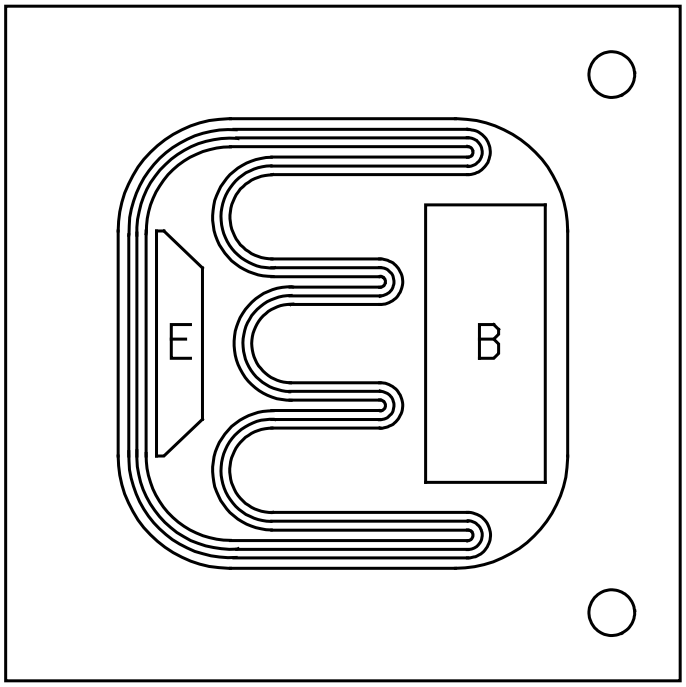

| Symbol | Dimensions | | | | Note |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| BH | .046 | .056 | 1.17 | 1.42 | |
| BL | .115 | .128 | 2.92 | 3.25 | |
| BW | .085 | .108 | 2.16 | 2.74 | |
| CL | | .128 | | 3.25 | |
| CW | | .108 | | 2.74 | |
| LL1 | .022 | .038 | 0.56 | 0.96 | |
| LL2 | .017 | .035 | 0.43 | 0.89 | |
| LS1 | .036 | .040 | 0.91 | 1.02 | |
| LS2 | .071 | .079 | 1.81 | 2.01 | |
| LW | .016 | .024 | 0.41 | 0.61 | |
| r | | .008 | | .203 | |
| r1 | | .012 | | .305 | |
| r2 | | .022 | | .559 | |

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Hatched areas on package denote metallized areas.
4. Pad 1 = Base, Pad 2 = Emitter, Pad 3 = Collector, Pad 4 = Shielding connected to the lid.
5. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

* FIGURE 3. Physical dimensions, surface mount (2N720AUB).



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| | |
|---|---|
| Die size: | .030 x .030 inch (0.76 mm x 0.76 mm). |
| Die thickness: | .008 ±.0016 inch (0.20 mm ±0.041 mm). |
| Base pad | .004 x .010 inch (0.10 mm x 0.254 mm). |
| Emitter pad: | .0023 x .007 inch (0.058 mm x 0.18 mm). |
| Back metal: | Gold, 6.5 kÅ ±1.95 kÅ. |
| Top metal: | Aluminum, 12 kÅ. Minimum; 14.5 kÅ. nominal. |
| Back side: | Collector. |
| Glassivation: | $SiO_2$, 7.5 kÅ ±1.5 kÅ. |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
* 3. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

* FIGURE 4. JANHCA2N720A and JANKCA2N720A die dimensions.



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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

| | |
|---|---|
| $R_{\theta JA}$ | Thermal resistance junction to ambient. |
| $R_{\theta JC}$ | Thermal resistance junction to case. |
| UB | Surface mount case outlines (see figure 3). |

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figures 1, 2, 3, and 4.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.6 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.7 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).



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4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.2.2 JANHC and JANKC qualification. JANHC and JANKC qualification inspection shall be in accordance with MIL-PRF-19500.

4.3 Screening (list applicable JANTX and JANTXV levels). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see table E-IV of MIL-PRF-19500) | Measurement |
|---|---|
| | JANTX and JANTXV levels |
| (1) 3a | Thermal impedance (see 4.3.2). |
| 7 | Optional |
| 9 | Not applicable |
| 11 | $I_{CBO2}$, $h_{FE3}$ |
| 12 | See 4.3.1. |
| 13 | $\Delta I_{CBO2}$ = 100 percent of initial value or 5 nA dc, whichever is greater; $\Delta h_{FE3}$ = 15 percent of initial value, subgroup 2 of table I herein. |
| 14 | Required |

(1) Shall be performed anytime after temperature cycling, screen 3a; and does not need to be repeated in screening requirements.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: $V_{CB}$ = 10 - 30 V dc. Power shall be applied to achieve $T_J$ = +135°C minimum using a minimum $P_D$ = 75 percent of $P_T$ maximum rated as defined in 1.3. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, $T_J$, and mounting conditions) may be used for JANTX and JANTXV quality levels. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval.

4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining $I_M$, $I_H$, $t_H$, $t_{SW}$ ($V_C$ and $V_H$ where appropriate). Measurement delay time ($t_{MD}$) = 70 μs max. See group E, subgroup 4 herein.

4.3.3 Screening (JANHC and JANKC). Screening of JANHC and JANKC die shall be in accordance with MIL-PRF-19500, "Discrete Semiconductor Die/Chip Lot Acceptance". As a minimum, die shall be 100-percent probed to ensure compliance with group A, subgroup 2. Burn-in duration for the JANKC level follows JANS requirements; the JANHC follows JANTX requirements.



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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. See 4.4.2.1 for JAN, JANTX, and JANTXV group B testing. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.4.2.1 Group B inspection, (JANJ, JANTX, and JANTXV). Separate samples may be used for each step. In the event of a lot failure, the resubmission requirements of MIL-PRF-19500 shall apply. In addition, all catastrophic failures during CI shall be analyzed to the extent possible to identify root cause and corrective action.

| Step | Method | Condition |
|---|---|---|
| 1 | 1026 | Steady-state life: 1,000 hours minimum, $V_{CB}$ = 10 V dc, power shall be applied to achieve $T_J$ = +150°C minimum using a minimum of $P_D$ = 75 percent of maximum rated $P_T$ as defined in 1.3. n = 45 devices, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours. |
| 2 | 1048 | Blocking life, $T_A$ = 150°C, $V_{CB}$ = 80 percent of rated voltage, 48 hours minimum. n = 45 devices, c = 0. |
| 3 | 1032 | High-temperature life (non-operating), t = 340 hours $T_A$ = +200°C. n = 22, c = 0. |

4.4.2.2 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. For JAN, JANTX, and JANTXV samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot.

b. Shall be chosen from an inspection lot that has been submitted to and passed table I, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

* 4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and 4.4.3.1 (JAN, JANTX, JANTXV) herein for group C testing. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

| | Subgroup | Method | Condition |
|---|---|---|---|
| | C2 | 2036 | Test condition E (not applicable to the 2N720AUB). |
| * | C5 | 3131 | $R_{\theta JA}$ and $R_{\theta JC}$ only, as applicable (see 1.3 and 4.3.2). |
| | C6 | 1026 | Not applicable. |

4.4.3.1 Group C sample selection. Samples selected from group C inspection shall be chosen at random from any lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes group A tests for conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for C6 life test may be pulled prior to the application of final lead finish. Testing of a group using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.



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4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table II herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.5 Method of inspection. Methods of inspection shall be as specified in appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.



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* TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 2/ | | | | | | |
| Visual and mechanical examination 3/ | 2071 | n = 45 devices, c = 0 | | | | |
| Solderability 3/ | 2026 | n = 15 leads, c = 0 | | | | |
| Resistance to solvents 3/ 4/ | 1022 | n = 15 devices, c = 0 | | | | |
| Temp cycling 3/ | 1051 | Test condition C, 25 cycles. n = 22 devices, c = 0 | | | | |
| Hermetic seal 5/<br>Fine leak<br>Gross leak | 1071 | n = 22 devices, c = 0 | | | | |
| Electrical measurements | | Table I, subgroup 2 | | | | |
| Bond strength 3/ | 2037 | Precondition $T_A$ = +250°C at t = 24 hours or $T_A$ = +300°C at t = 2 hours, n = 11 wires, c = 0 | | | | |
| Decap internal visual (design verification) | 2075 | n = 4 devices, c = 0 | | | | |
| Subgroup 2 | | | | | | |
| Thermal impedance | 3131 | See 4.3.2 | $Z_{\theta JX}$ | | | °C/W |
| Collector to base cutoff current | 3036 | Bias condition D, $V_{CB}$ = 120 V dc pulsed (see 4.5.1). | $I_{CBO1}$ | | 100 | µA dc |
| Emitter to base cutoff current | 3061 | Bias condition D, $V_{EB}$ = 7 V dc pulsed (see 4.5.1). | $I_{EBO1}$ | | 100 | µA dc |
| Breakdown voltage, collector-emitter | 3011 | Bias condition D; $I_C$ = 30 mA dc; pulsed (see 4.5.1) | $V_{(BR)CEO}$ | 80 | | V dc |
| Breakdown voltage collector-emitter | 3011 | Bias condition D, $I_C$ = 10 mA dc, pulsed (see 4.5.1), $R_{BE}$ = 10 Ω | $V_{(BR)CER}$ | 100 | | V dc |
| Collector to base cutoff current | 3036 | Bias condition D, $V_{CB}$ = 90 V dc | $I_{CBO2}$ | | 10 | nA dc |
| Emitter-base cutoff current | 3061 | Bias condition D, $V_{EB}$ = 5 V dc | $I_{EBO2}$ | | 10 | nA dc |
| Collector-emitter saturated voltage | 3071 | $I_C$ = 150 mA dc, $I_B$ = 15 mA dc, pulsed (see 4.5.1) | $V_{CE(sat)1}$ | | 5.0 | V dc |

See footnotes at end of table.



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* TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 2 - Continued | | | | | | |
| Collector-emitter saturated voltage | 3071 | $I_C$ = 50 mA dc, $I_B$ = 5 mA dc, pulsed (see 4.5.1) | $V_{CE(sat)2}$ | | 1.2 | V dc |
| Base-emitter saturated voltage | 3066 | Test condition A, $I_C$ = 150 mA dc, $I_B$ = 15 mA dc, pulsed (see 4.5.1) | $V_{BE(sat)1}$ | | 1.3 | V dc |
| Base-emitter saturated voltage | 3066 | Test condition A, $I_C$ = 50 mA dc, $I_B$ = 5 mA dc, pulsed (see 4.5.1) | $V_{BE(sat)2}$ | | 1.2 | V dc |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc, $I_C$ = 0.1 mA dc, pulsed (see 4.5.1) | $h_{FE1}$ | 20 | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc, $I_C$ = 10 mA dc, pulsed (see 4.5.1) | $h_{FE2}$ | 35 | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc, $I_C$ = 150 mA dc, pulsed (see 4.5.1) | $h_{FE3}$ | 40 | 120 | |
| Subgroup 3 | | | | | | |
| High temperature operation: | | $T_A$ = +150°C | | | | |
| Collector to base cutoff current | 3036 | Bias condition D, $V_{CB}$ = 90 V dc | $I_{CBO3}$ | | 15 | µA dc |
| Low temperature operation | | $T_A$ = -55°C | | | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc, $I_C$ = 10 mA dc, pulsed (see 4.5.1) | $h_{FE4}$ | 20 | | |
| Subgroup 4 | | | | | | |
| Small signal short circuit forward current transfer ratio | 3206 | $V_{CE}$ = 5 V dc, f = 1 kHz, $I_C$ = 1 mA dc: | $h_{fe}$ | 35 | 100 | |
| | | $V_{CE}$ = 10 V dc, f = 1 kHz, $I_C$ = 5 mA dc | $h_{fe}$ | 45 | 150 | |
| Magnitude of common emitter small-signal short-circuit forward-current transfer ratio | 3306 | $V_{CE}$ = 10 V dc, f = 20 MHz, $I_C$ = 50 mA dc | $\lvert h_{fe} \rvert$ | 3 | 10 | |
| Small signal short circuit input impedance | 3201 | $V_{CB}$ = 10 V dc, $I_C$ = 5 mA dc | $h_{ib}$ | 4 | 8 | ohms |

See footnotes at end of table.



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* TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 4 - Continued | | | | | | |
| Small signal short circuit output admittance | 3216 | $V_{CB}$ = 10 V dc, $I_C$ = 5 mA dc | $h_{ob}$ | 0 | 0.5 | µ ohms |
| Small signal open circuit reverse voltage transfer ratio | 3211 | $V_{CB}$ = 10 V dc, $I_C$ = 5 mA dc | $h_{rb}$ | | $1.5 \times 10^{-4}$ | |
| Open circuit output capacitance | 3236 | $V_{CB}$ = 10 V dc, $I_E$ = 0, 100 kHz ≤ f ≤ 1 MHz | $C_{obo}$ | 2 | 15 | pF |
| Pulse response | 3251 | Test condition A, except test circuit and pulse requirements. See figure 9 herein. | $t_{on} + t_{off}$ | | 30 | ns |
| Subgroups 5, 6, and 7 | | | | | | |
| Not appicable | | | | | | |

1/ For sampling plan see MIL-PRF-19500.
2/ For resubmission of failed in subgroup 1 of table I, double the sample size of the failed test or sequence of tests. A failure in table I, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for laser marked devices.
5/ This hermetic seal test is an end-point to temp-cycling in addition to electrical measurements.



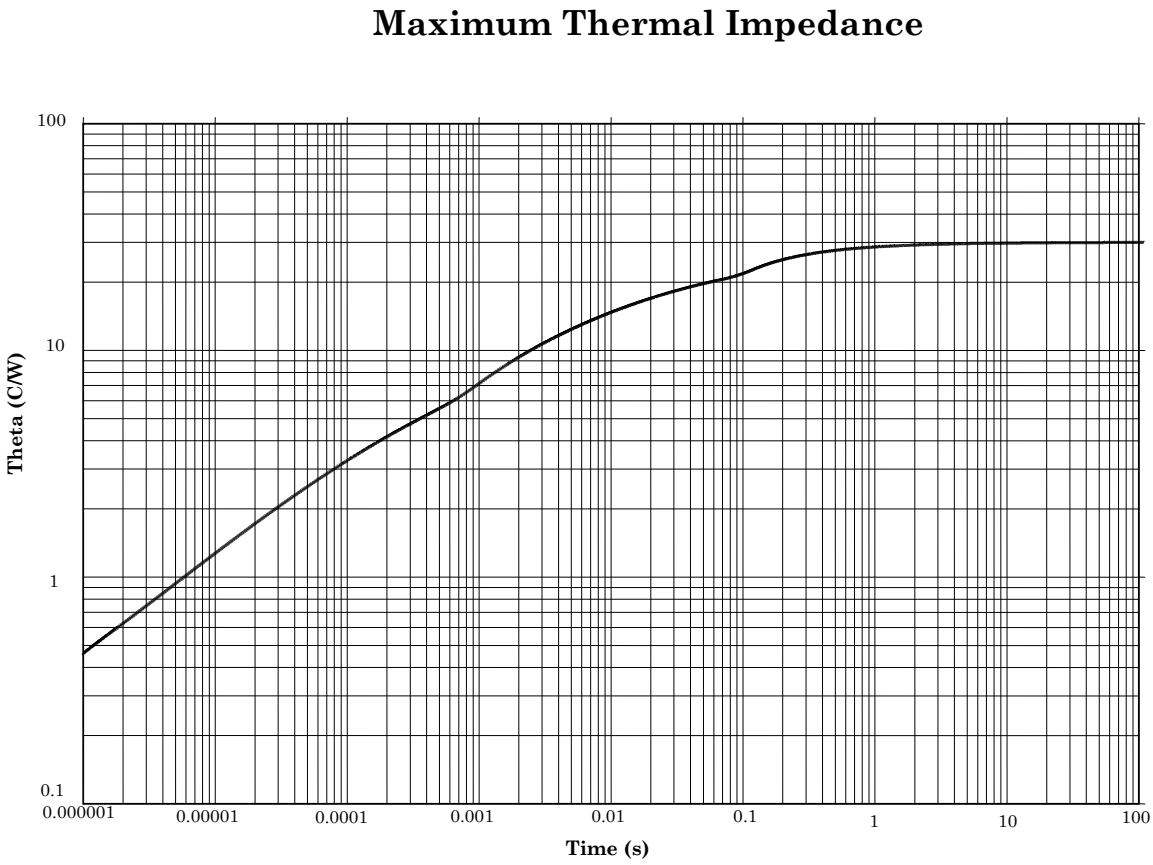
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* TABLE II. Group E inspection (all quality levels) - for qualification or re-qualification only.

| Inspection | MIL-STD-750 | | Qualification |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 45 devices<br>c = 0 |
| Temperature cycling (air to air) | 1051 | Test condition C, 500 cycles | |
| Hermetic seal | 1071 | | |
| Fine leak<br>Gross leak | | | |
| Electrical measurements | | See table I, subgroup 2 herein. | |
| Subgroup 2 | | | 45 devices<br>c = 0 |
| Intermittent life | 1037 | Intermittent operation life: $V_{CB}$ = 10 V dc, 6,000 cycles. Adjust device current, or power, to achieve a minimum $\Delta T_J$ of +100°C. | |
| Electrical measurements | | See table I, subgroup 2 herein. | |
| Subgroup 4 | | | |
| Thermal resistance | 3131 | $R_{\theta JSP(IS)}$ can be calculated but shall be measured once in the same package with a similar die size to confirm calculations (may apply to multiple slash sheets). | 15 devices,<br>c = 0 |
| Thermal impedance curves. | | See table E-IX of MIL-PRF-19500, group E, subgroup 4. | Sample size<br>N/A |
| Subgroup 5 | | | |
| Not applicable | | | |
| Subgroup 6 | | | 3 devices |
| Electrocstatic discharge (ESD) | 1020 | | |
| Subgroup 8 | | | 45 devices<br>c = 0 |
| Reverse stability | 1033 | Condition B. | |



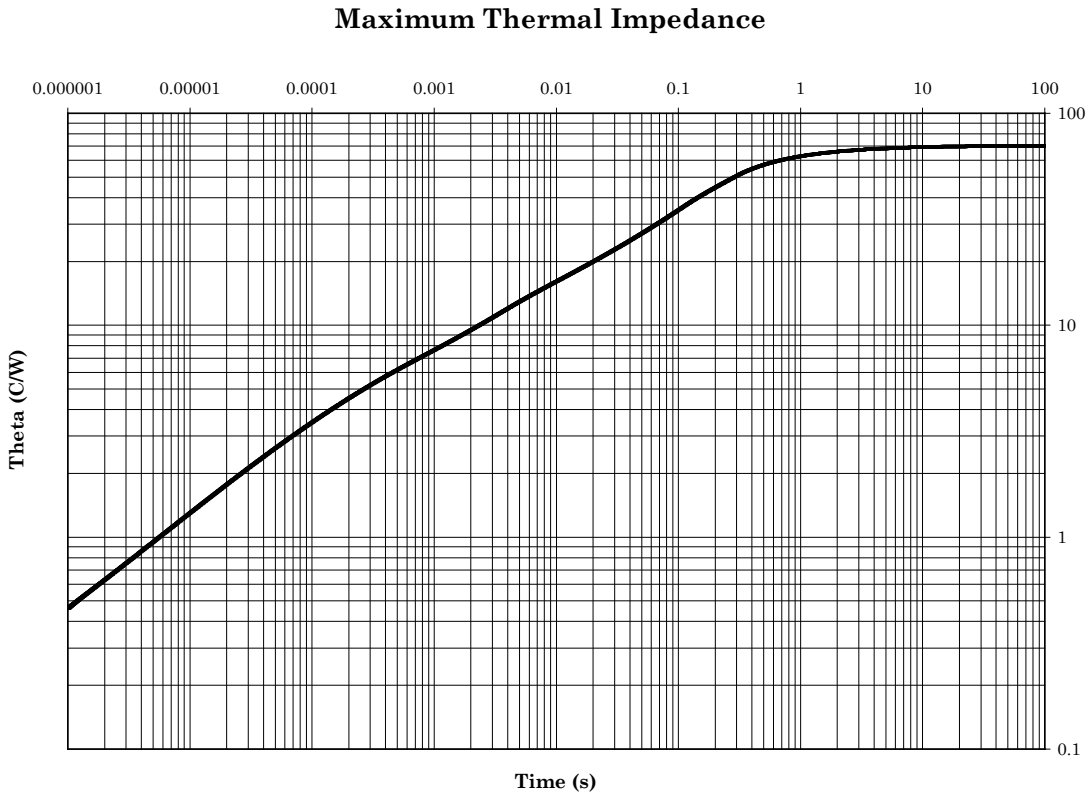
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* FIGURE 5. Thermal impedance graph ($R_{\theta JC}$) for 2N1893 and 2N1893S (TO-5).



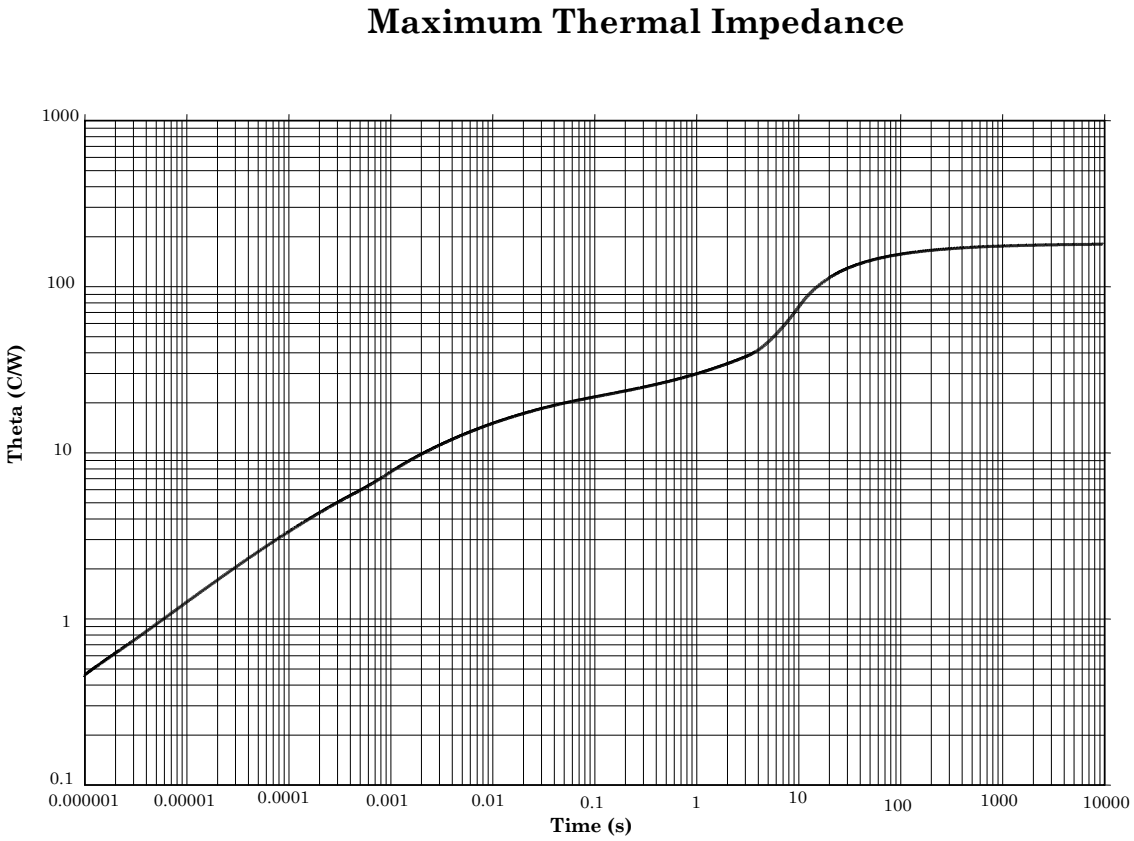
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* FIGURE 6. Thermal impedance graph ($R_{\theta JC}$) for 2N1893 and 2N1893S (TO-18).



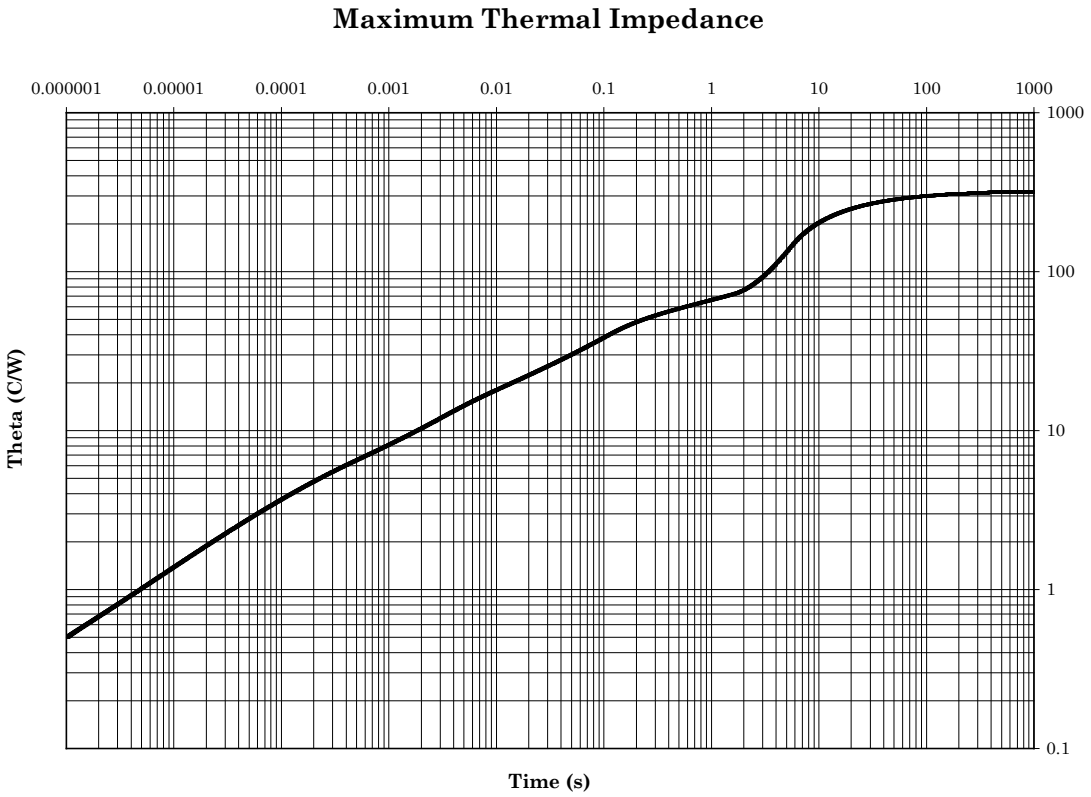
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* FIGURE 7. Thermal impedance graph ($R_{\theta JA}$) for 2N1893 and 2N1893S (TO-5).



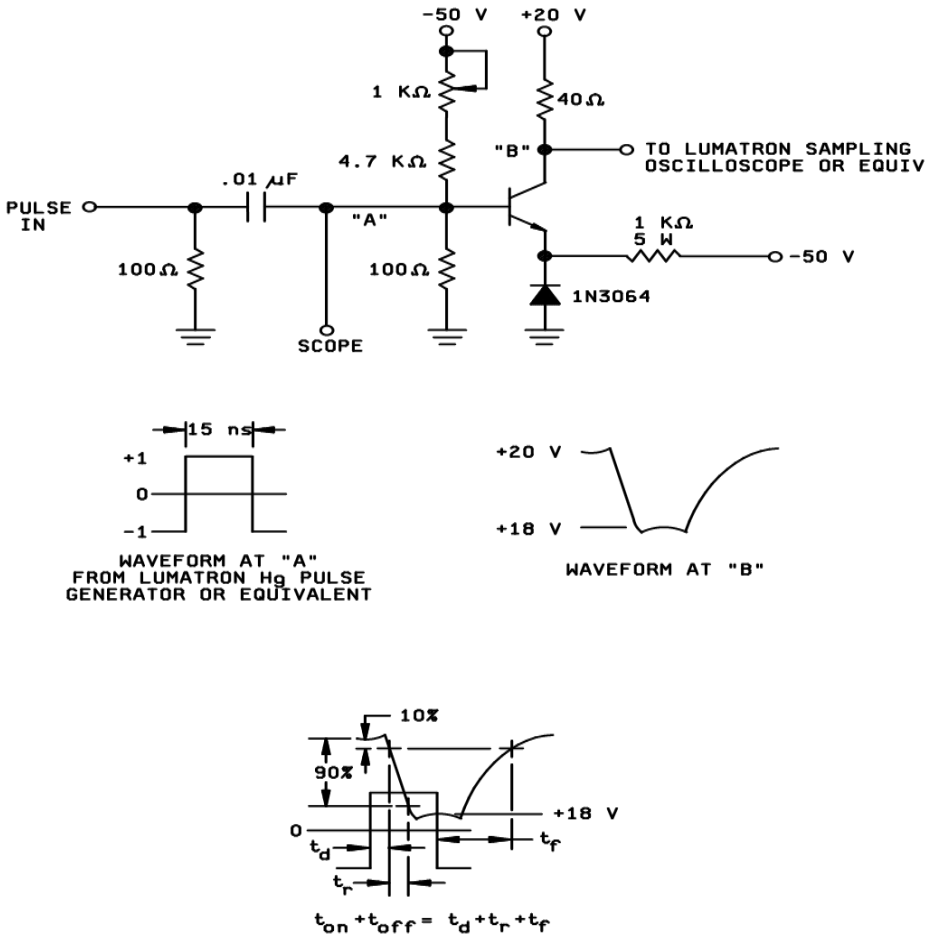
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* FIGURE 8. Thermal impedance graph ($R_{\theta JA}$) for 2N1893 and 2N1893S (TO-18).



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NOTES:

1. The rise time ($T_r$) of the applied pulse shall be 2.0 ≤ nanoseconds, duty cycle ≤ 2 percent, and the generator source impedance shall be 50 ohms.
2. Sampling oscilloscope: $Z_{in}$ 100 K ohms, $C_{in}$ ≤ 12 pF, rise time ≤ 5 nsec.

* FIGURE 9. Pulse response (turn-on plus turn-off) measurement circuit and waveforms.



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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

* (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

* 6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.daps.dla.mil .

6.4 Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example JANHCA2N720A) will be identified on the QML

| Die ordering information | |
|---|---|
| PIN | Manufacturer |
| | 34156 |
| 2N720A | JANHCA2N720A<br>JANKCA2N720A |



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* 6.5 Amendment notaions. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Review activities:
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Preparing activity:
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